Notebook

Exported 2026-11-11 · 14 cells

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option("display.max_columns", None)
pd.set_option("display.width", 120)

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
```

```
path = r"C:\Users\Surender\Downloads\train.csv"
df = pd.read_csv(path)
df.head()
```

Columns 1–8 of 13

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0

Columns 9–13 of 13 · first column repeated

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
print("Shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())

print("\nInfo:")
df.info()

print("\nDescribe:")
display(df.describe(include="all").T)

print("\nMissing values:")
missing = df.isna().sum().sort_values(ascending=False)
display(pd.DataFrame({
    "missing_count": missing,
    "missing_percent": (missing / len(df) * 100).round(2)
}))
```

```
Shape: (891, 12)

Columns:
['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket',
'Fare', 'Cabin', 'Embarked']

Info:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

Describe:
```

Columns 1–8 of 12

	count	unique	top	freq	mean	std	min
PassengerId	891.0	NaN	NaN	NaN	446.0	257.353842	1.0
Survived	891.0	NaN	NaN	NaN	0.383838	0.486592	0.0
Pclass	891.0	NaN	NaN	NaN	2.308642	0.836071	1.0
Name	891	891	Braund, Mr. Owen Harris	1	NaN	NaN	NaN

	count	unique	top	freq	mean	std	min
Sex	891	2	male	577	NaN	NaN	NaN
Age	714.0	NaN	NaN	NaN	29.699118	14.526497	0.42
SibSp	891.0	NaN	NaN	NaN	0.523008	1.102743	0.0
Parch	891.0	NaN	NaN	NaN	0.381594	0.806057	0.0
Ticket	891	681	347082	7	NaN	NaN	NaN
Fare	891.0	NaN	NaN	NaN	32.204208	49.693429	0.0
Cabin	204	147	G6	4	NaN	NaN	NaN
Embarked	889	3	S	644	NaN	NaN	NaN

Columns 9–12 of 12 · first column repeated

	25%	50%	75%	max
PassengerId	223.5	446.0	668.5	891.0
Survived	0.0	0.0	1.0	1.0
Pclass	2.0	3.0	3.0	3.0
Name	NaN	NaN	NaN	NaN
Sex	NaN	NaN	NaN	NaN
Age	20.125	28.0	38.0	80.0
SibSp	0.0	0.0	1.0	8.0
Parch	0.0	0.0	0.0	6.0
Ticket	NaN	NaN	NaN	NaN
Fare	7.9104	14.4542	31.0	512.3292
Cabin	NaN	NaN	NaN	NaN
Embarked	NaN	NaN	NaN	NaN

```
Missing values:
```

	missing_count	missing_percent
Cabin	687	77.10
Age	177	19.87
Embarked	2	0.22
PassengerId	0	0.00
Name	0	0.00
Pclass	0	0.00
Survived	0	0.00
Sex	0	0.00
Parch	0	0.00
SibSp	0	0.00
Fare	0	0.00
Ticket	0	0.00

```
for col in ["Survived", "Pclass", "Sex", "Embarked"]:
    print(f"\nValue counts for {col}:")
    display(df[col].value_counts(dropna=False))

print("\nSurvival rate:")
print(df["Survived"].value_counts(normalize=True).round(4))
```

```
Value counts for Survived:
```

```
Survived
0    549
1    342
Name: count, dtype: int64
```

```
Value counts for Pclass:
```

```
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

```
Value counts for Sex:
```

```
Sex
male      577
female    314
Name: count, dtype: int64
```

```
Value counts for Embarked:
```

```
Embarked
S      644
C      168
Q       77
NaN      2
Name: count, dtype: int64
```

```
Survival rate:
Survived
0    0.6162
1    0.3838
Name: proportion, dtype: float64
```

```
df_eda = df.copy()

df_eda["FamilySize"] = df_eda["SibSp"] + df_eda["Parch"] + 1
df_eda["IsAlone"] = (df_eda["FamilySize"] == 1).astype(int)

df_eda["AgeGroup"] = pd.cut(
    df_eda["Age"],
    bins=[0, 12, 18, 35, 60, 100],
    labels=["Child", "Teen", "Adult", "MiddleAge", "Senior"]
)

df_eda["FareGroup"] = pd.qcut(
    df_eda["Fare"],
    q=4,
    labels=["Low", "MidLow", "MidHigh", "High"],
    duplicates="drop"
)

df_eda["Sex_male"] = (df_eda["Sex"] == "male").astype(int)

df_eda.head()
```

Columns 1–8 of 18

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0

Columns 9–15 of 18 · first column repeated

	Parch	Ticket	Fare	Cabin	Embarked	FamilySize	IsAlone
0	0	A/5 21171	7.2500	NaN	S	2	0
1	0	PC 17599	71.2833	C85	C	2	0
2	0	STON/O2. 3101282	7.9250	NaN	S	1	1
3	0	113803	53.1000	C123	S	2	0
4	0	373450	8.0500	NaN	S	1	1

Columns 16–18 of 18 · first column repeated

	AgeGroup	FareGroup	Sex_male
0	Adult	Low	1
1	MiddleAge	High	0
2	Adult	MidLow	0
3	Adult	High	0
4	Adult	MidLow	1

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

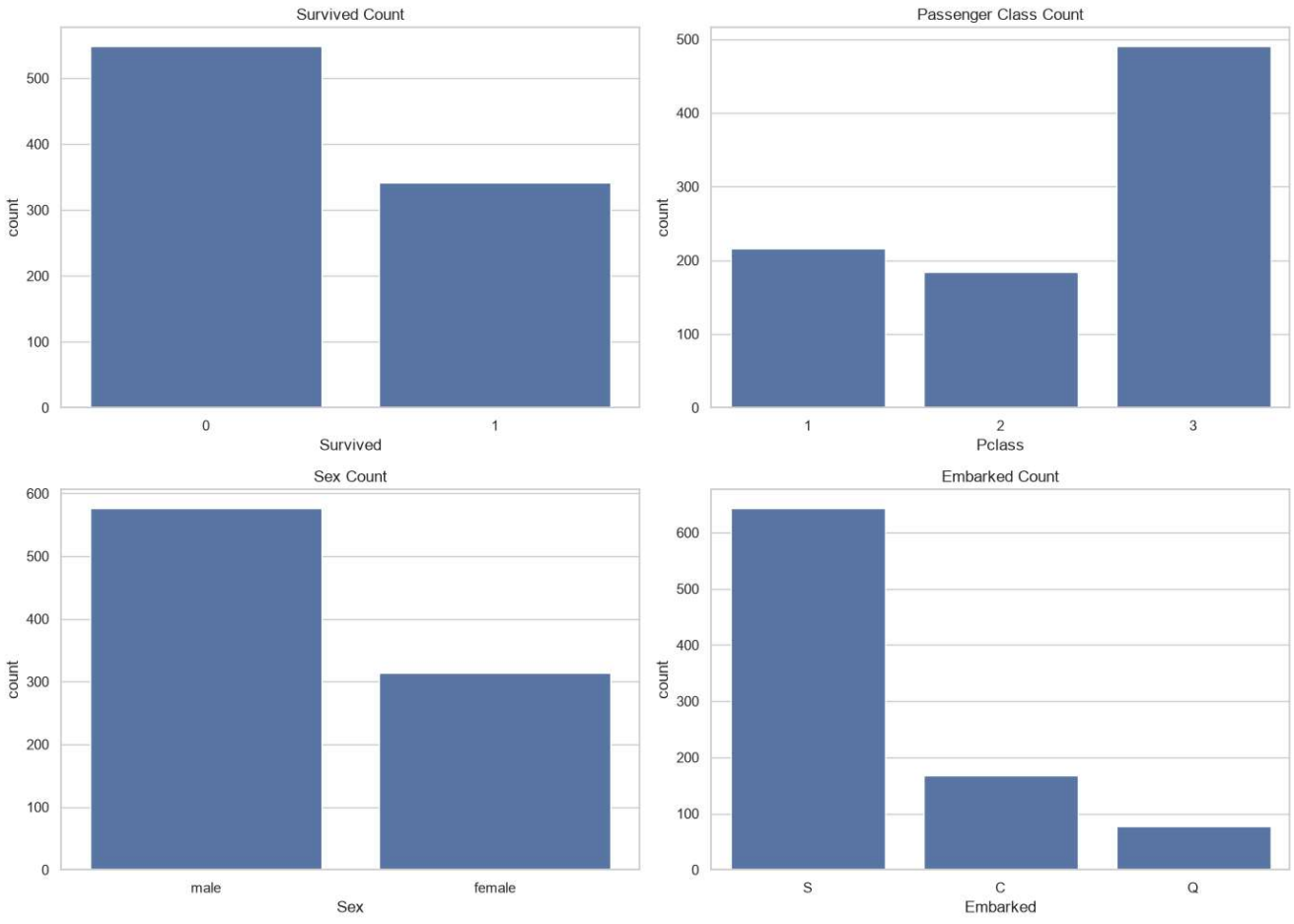
sns.countplot(data=df_eda, x="Survived", ax=axes[0, 0])
axes[0, 0].set_title("Survived Count")

sns.countplot(data=df_eda, x="Pclass", ax=axes[0, 1])
axes[0, 1].set_title("Passenger Class Count")

sns.countplot(data=df_eda, x="Sex", ax=axes[1, 0])
axes[1, 0].set_title("Sex Count")

sns.countplot(data=df_eda, x="Embarked", ax=axes[1, 1])
axes[1, 1].set_title("Embarked Count")

plt.tight_layout()
plt.show()
```

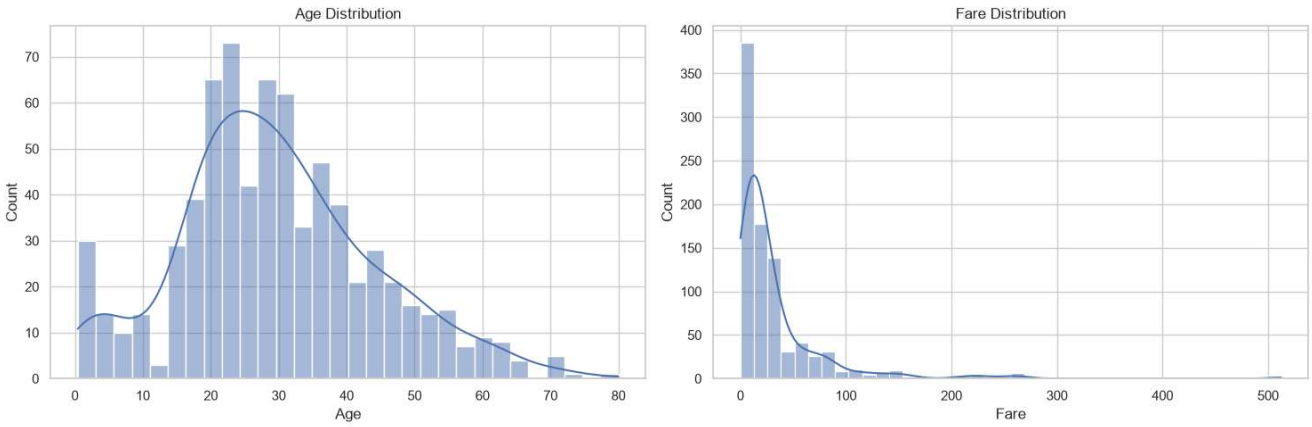


```
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

sns.histplot(df_eda["Age"].dropna(), bins=30, kde=True, ax=axes[0])
axes[0].set_title("Age Distribution")

sns.histplot(df_eda["Fare"], bins=40, kde=True, ax=axes[1])
axes[1].set_title("Fare Distribution")

plt.tight_layout()
plt.show()
```



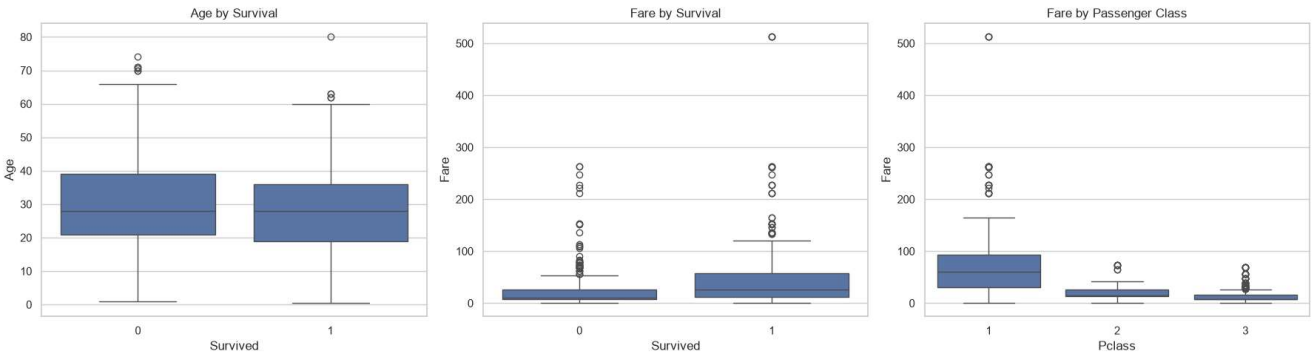

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

sns.boxplot(data=df_eda, x="Survived", y="Age", ax=axes[0])
axes[0].set_title("Age by Survival")

sns.boxplot(data=df_eda, x="Survived", y="Fare", ax=axes[1])
axes[1].set_title("Fare by Survival")

sns.boxplot(data=df_eda, x="Pclass", y="Fare", ax=axes[2])
axes[2].set_title("Fare by Passenger Class")

plt.tight_layout()
plt.show()
```



```
def survival_table(col):
    return (
        df_eda.groupby(col, dropna=False)["Survived"]
        .agg(["count", "mean"])
        .rename(columns={"mean": "survival_rate"})
        .assign(survival_rate=lambda x: (x["survival_rate"] * 100).round(2))
        .sort_values("survival_rate", ascending=False)
    )

print("Survival by Sex:")
display(survival_table("Sex"))

print("\nSurvival by Pclass:")
display(survival_table("Pclass"))

print("\nSurvival by Embarked:")
display(survival_table("Embarked"))

print("\nSurvival by FamilySize:")
display(survival_table("FamilySize"))

print("\nSurvival by AgeGroup:")
display(survival_table("AgeGroup"))

print("\nSurvival by FareGroup:")
display(survival_table("FareGroup"))
```

```
Survival by Sex:
```

	count	survival_rate
Sex		
female	314	74.20
male	577	18.89

```
Survival by Pclass:
```

	count	survival_rate
Pclass		
1	216	62.96
2	184	47.28
3	491	24.24

```
Survival by Embarked:
```

	count	survival_rate
Embarked		
NaN	2	100.00
C	168	55.36
Q	77	38.96
S	644	33.70

```
Survival by FamilySize:
```

FamilySize	count	survival_rate
4	29	72.41
3	102	57.84
2	161	55.28
7	12	33.33
1	537	30.35
5	15	20.00
6	22	13.64
8	6	0.00
11	7	0.00

```
Survival by AgeGroup:
```

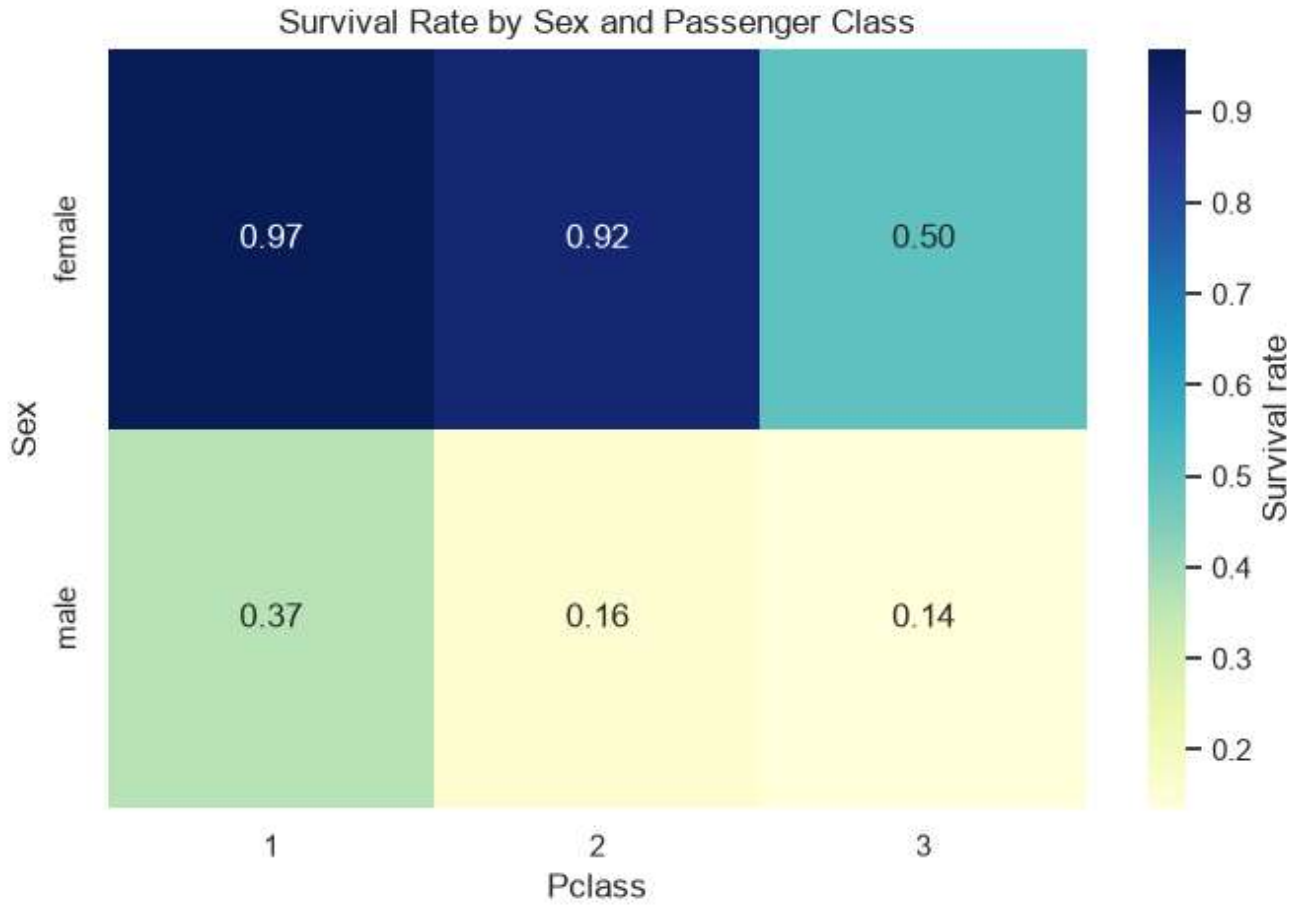
AgeGroup	count	survival_rate
Child	69	57.97
Teen	70	42.86
MiddleAge	195	40.00
Adult	358	38.27
NaN	177	29.38
Senior	22	22.73

```
Survival by FareGroup:
```

FareGroup	count	survival_rate
High	222	58.11
MidHigh	222	45.50
MidLow	224	30.36
Low	223	19.73

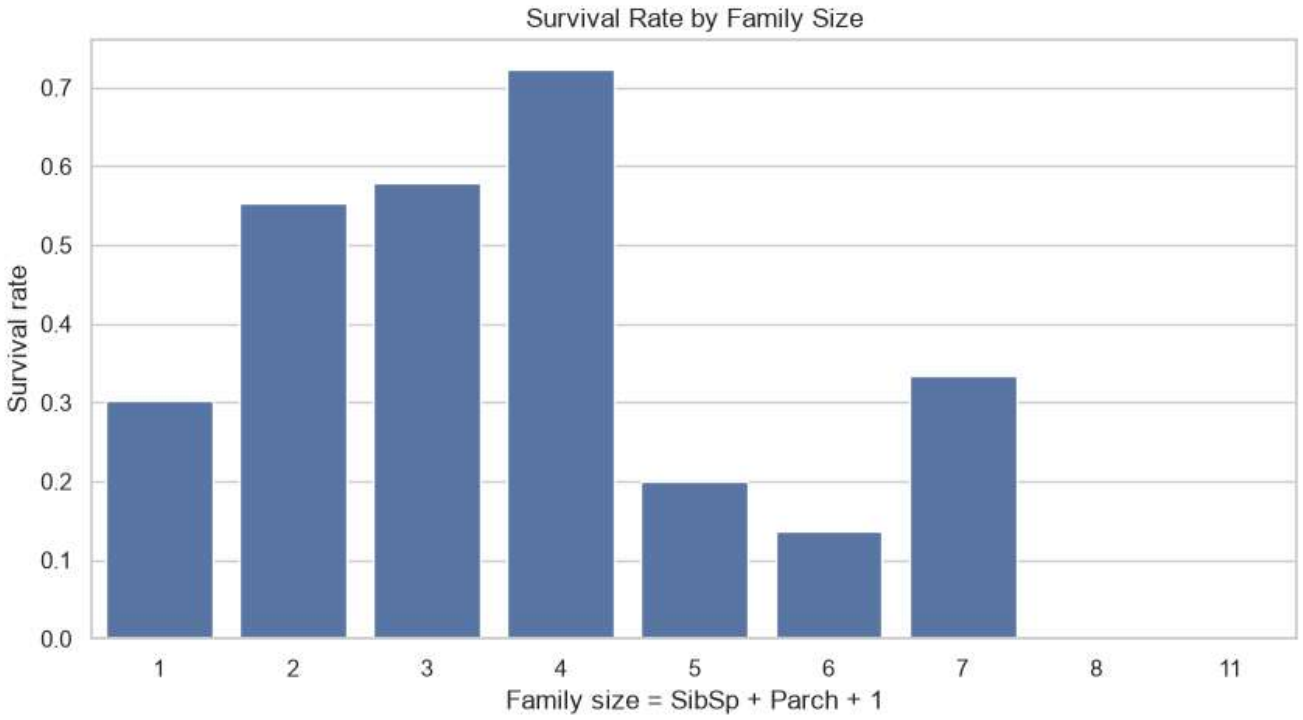
```
pivot_sex_class = df_eda.pivot_table(
    values="Survived",
    index="Sex",
    columns="Pclass",
    aggfunc="mean"
)

plt.figure(figsize=(8, 5))
sns.heatmap(
    pivot_sex_class,
    annot=True,
    fmt=".2f",
    cmap="YlGnBu",
    cbar_kws={"label": "Survival rate"}
)
plt.title("Survival Rate by Sex and Passenger Class")
plt.show()
```



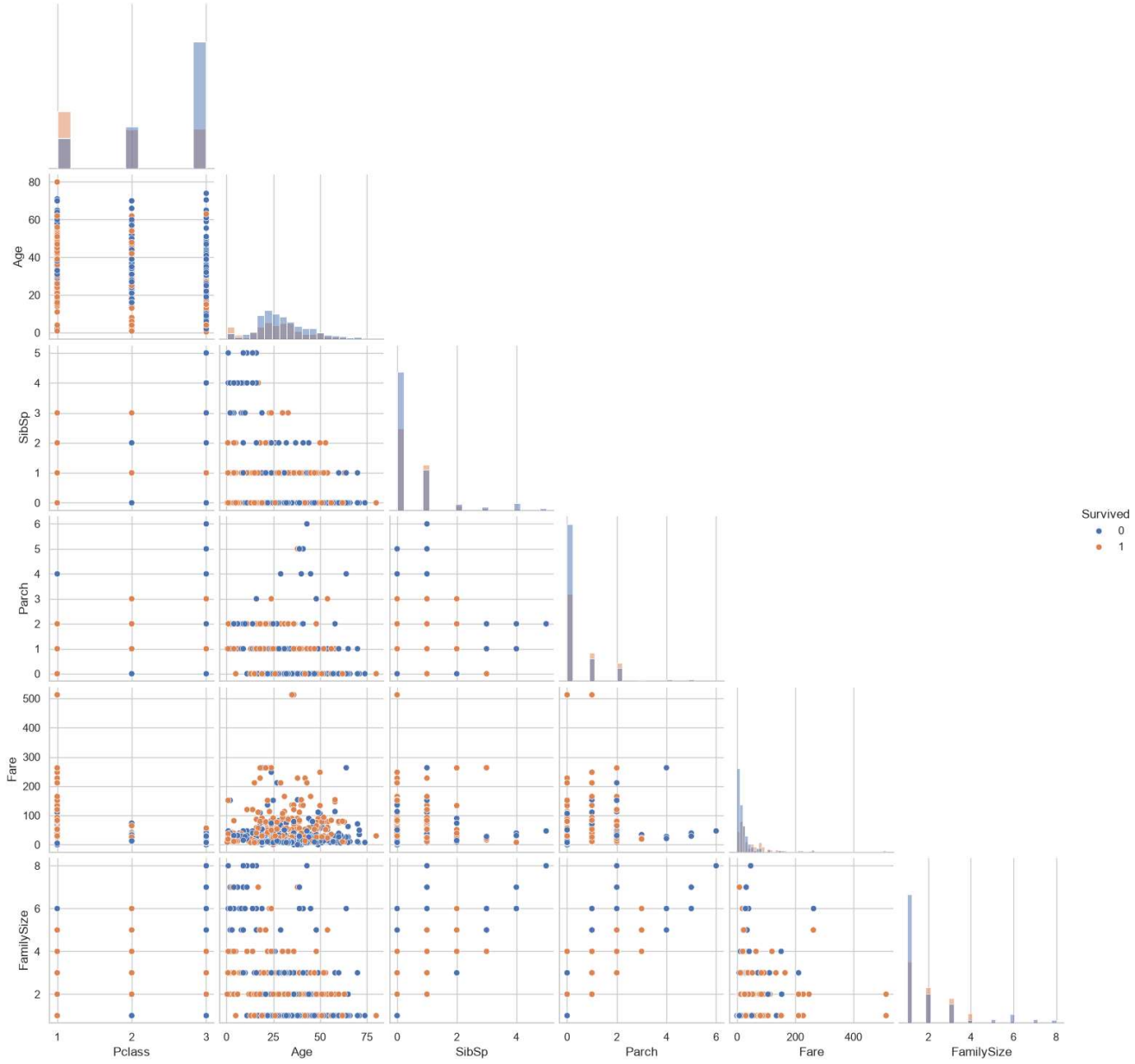
```
family_survival = (
    df_eda.groupby("FamilySize")["Survived"]
    .agg(["count", "mean"])
    .reset_index()
)

plt.figure(figsize=(10, 5))
sns.barplot(data=family_survival, x="FamilySize", y="mean")
plt.title("Survival Rate by Family Size")
plt.ylabel("Survival rate")
plt.xlabel("Family size = SibSp + Parch + 1")
plt.show()
```



```
numeric_cols = ["Survived", "Pclass", "Age", "SibSp", "Parch", "Fare", "FamilySize"]

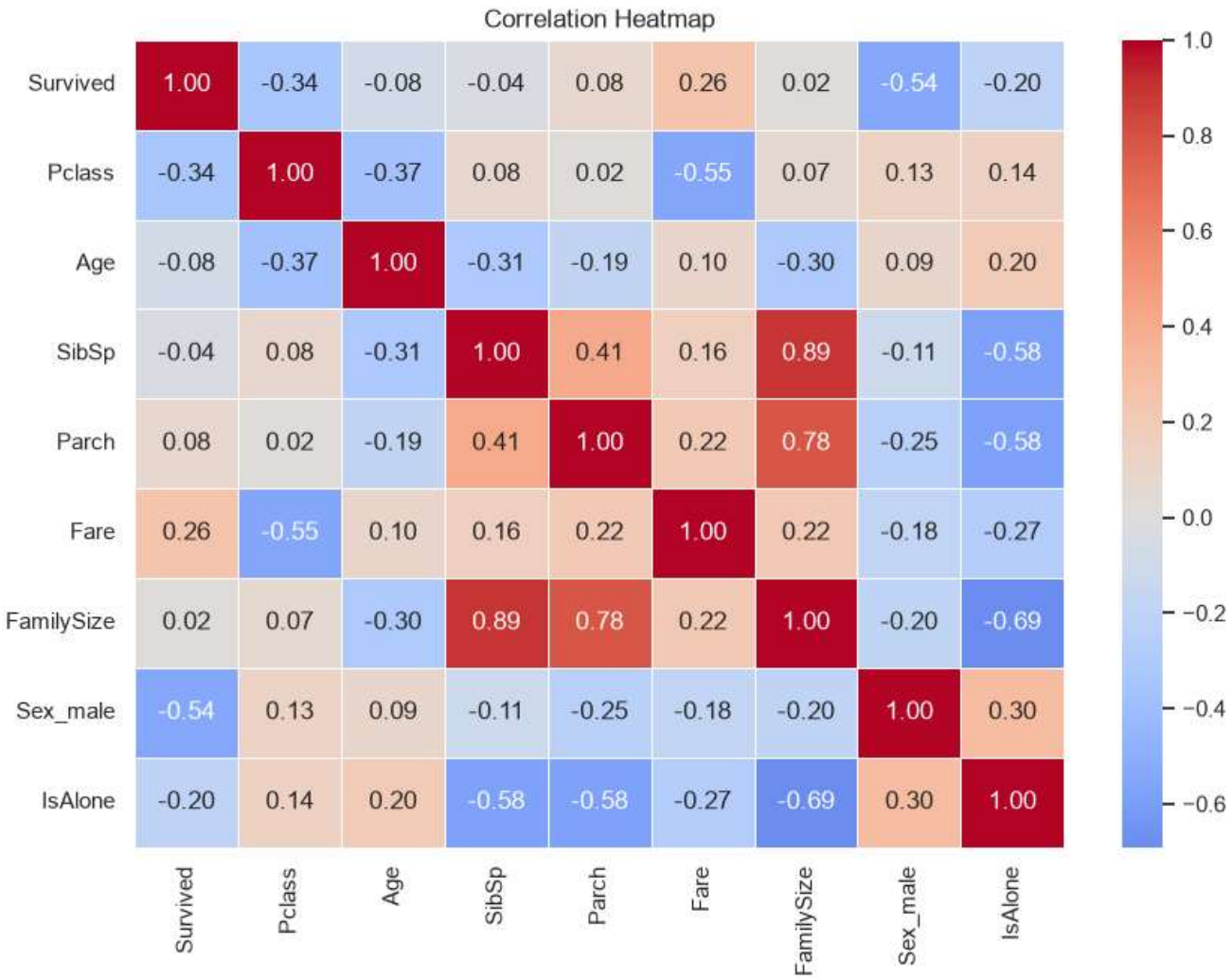
sns.pairplot(
    df_eda[numeric_cols].dropna(),
    hue="Survived",
    diag_kind="hist",
    corner=True
)
plt.show()
```



```
corr_cols = [
    "Survived", "Pclass", "Age", "SibSp", "Parch",
    "Fare", "FamilySize", "Sex_male", "IsAlone"
]

corr = df_eda[corr_cols].corr()

plt.figure(figsize=(10, 7))
sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0,
    linewidths=0.5
)
plt.title("Correlation Heatmap")
plt.show()
```



```
# Fare outliers using IQR
Q1 = df_eda["Fare"].quantile(0.25)
Q3 = df_eda["Fare"].quantile(0.75)
IQR = Q3 - Q1

fare_outliers = df_eda[(df_eda["Fare"] < Q1 - 1.5 * IQR) |
                       (df_eda["Fare"] > Q3 + 1.5 * IQR)]

print("Number of Fare outliers by IQR rule:", len(fare_outliers))
display(fare_outliers[["PassengerId", "Name", "Pclass", "Fare", "Survived"]].head(10))

print("Zero Fare rows:", (df_eda["Fare"] == 0).sum())
print("Max Fare:", df_eda["Fare"].max())
print("Median Fare:", df_eda["Fare"].median())
print("Mean Fare:", round(df_eda["Fare"].mean(), 2))
```

```
Number of Fare outliers by IQR rule: 116
```

	PassengerId	Name	Pclass	Fare	Survived
1	2	Cumings, Mrs. John Bradley (Florence Briggs Th...	1	71.2833	1
27	28	Fortune, Mr. Charles Alexander	1	263.0000	0
31	32	Spencer, Mrs. William Augustus (Marie Eugenie)	1	146.5208	1
34	35	Meyer, Mr. Edgar Joseph	1	82.1708	0
52	53	Harper, Mrs. Henry Sleeper (Myna Haxtun)	1	76.7292	1
61	62	Icard, Miss. Amelie	1	80.0000	1
62	63	Harris, Mr. Henry Birkhardt	1	83.4750	0
72	73	Hood, Mr. Ambrose Jr	2	73.5000	0
88	89	Fortune, Miss. Mabel Helen	1	263.0000	1
102	103	White, Mr. Richard Frasar	1	77.2875	0

```
Zero Fare rows: 15
Max Fare: 512.3292
Median Fare: 14.4542
Mean Fare: 32.2
```